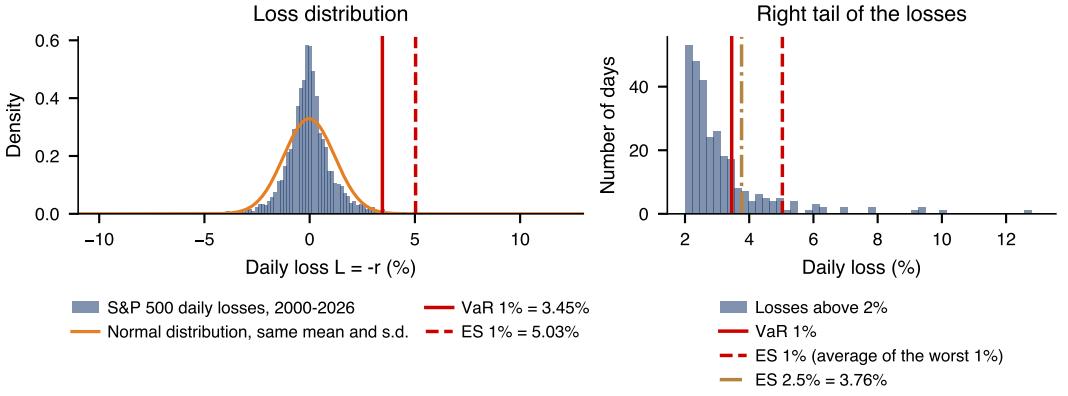
Loss distribution
Density
0.6
0.4
0.2
0.0
−10
−5
0
5
10
Daily loss L = -r (%)
S&P 500 daily losses, 2000-2026
Normal distribution, same mean and s.d.
VaR 1% = 3.45%
ES 1% = 5.03%
Right tail of the losses
Number of days
40
20
0
2
4
6
8
10
12
Daily loss (%)
Losses above 2%
VaR 1%
ES 1% (average of the worst 1%)
ES 2.5% = 3.76%